

# Masking a Field’s Loss Does Not Isolate That Field:

## Two Rejected LoRA Runs on a Structured-Output Guard Model

Anonymous Author(s)  
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### Abstract

Structured-output guard models emit several fields per inference, typically a severity verdict and a category. When one underperforms, a natural experiment is to mask its loss and tune the other, reasoning that withholding supervision preserves behavior.

It does not. We report two rejected fine-tuning runs on a 0.6B guard model. Run 1 supervised severity with the binary targets its corpus supports and extinguished its third severity class, 49 Controversial verdicts falling to 0, converting a documented operator lever into a no-op that still reported success. Run 2 masked the severity line in the loss while keeping it in the input, and scored 17 points of overall recall below run 1 and 23 adversarial points below the untuned base model. The masked field did not hold still: Controversial overproduced at 187 against a base of 49, Unsafe collapsed from 650 to 367, and the model began emitting its own category vocabulary in the severity slot.

Neither mechanism is new. Concurrent work reports a masked field degrading under the same configuration, and binary collapse is standing practice in guardrail dataset design. Three narrower findings survive: the per-class distribution of a masked categorical enum against base-model counts; the direction of the effect, opposite in sign to the concurrent result whose unmasked baseline supervises a degenerate target; and the governance consequence, that a masked field is a policy surface whose silent movement can disable an audited control.

We report both rejections, the automated gate’s two verdicts, and the corpus property that blocked the intended repair.

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## 1 Why report two runs that were rejected

Guard models are increasingly the primary runtime control in agentic deployments, and they are increasingly structured-output models rather than binary classifiers: a severity verdict *and* a category, parsed downstream by policy that may treat either as dispositive.

That structure invites a specific experiment. When one field is weak, tune the other and leave the first alone. The intuition is that supervision is what changes a model, so withholding supervision should preserve behavior.

We ran that experiment. It failed, and it failed in a direction that is worth more than a promotion would have been. Both runs were rejected by an automated parity gate before any promotion decision reached a human, and the gate’s two verdicts distinguish precisely between *no improvement demonstrated* and *this made it worse* — a distinction the literature

on fine-tuning guards does not usually get to make, because unsuccessful runs are rarely reported at all.

### Contributions.

**Novelty, bounded first.** That loss masking fails to isolate a field under a shared adapter is **documented concurrently**, and that binary targets produce binary behavior is **established practice**. §7 withdraws the broader claim an earlier draft made and cites the work that refutes it. Four narrower contributions survive:

1. **The distributional signature.** The per-class distribution of a masked *categorical enum* measured against the base model’s own counts — overproduction of one class, collapse of another, and migration of the neighboring field’s vocabulary into the masked slot (§4.2).
2. **The direction.** Masking cost 17 recall points *against supervising the same field with wrong targets*. That is the opposite sign to the concurrent result, whose unmasked baseline supervises a degenerate target, and the disagreement is informative rather than contradictory (§4.1, §7).
3. **The governance consequence.** A masked field is a policy surface. Its silent movement converted a documented operator control into a no-op that reported success (§3.3). We are aware of no treatment of masked-field drift as an auditability failure rather than a quality failure.
4. **The corpus constraint and the prescription.** The signal needed to supervise three-valued severity exists only where it cannot be trained on (§3.5), and isolation therefore requires separate parameters rather than a masking flag (§5).

## 2 Setup

**Model.** A 0.6B parameter generative guard model, fine-tuned with LoRA. A same-size untuned baseline is retained deliberately: without a same-size comparator, a rejection cannot distinguish *the fine-tune did not work* from *this model is smaller than the one it is being compared to* — different findings with different next actions.

**Output format.** Each inference emits a severity line and a category line. Severity is three-valued in the base model’s vocabulary — **Safe**, **Unsafe**, **Controversial**. Downstream policy treats a gating category as harmful in its own right, and separately allows an operator to decide whether **Controversial** counts as harmful.

**Corpus.** A public guard training mixture of 38,694 rows. **Its severity targets are binary:** the rendering function emits only **Safety: Unsafe** or **Safety: Safe**, and the corpus contains 19,349 of one and 19,345 of the other. There is no third value anywhere in the training targets.

**Evaluation.** The full held-out test split, 1,699 items, scored on recall first, with per-category recall and the confusion matrix reported alongside.

**Reported precision of every figure in this paper.** Headline rates are reported as recorded, to four decimal places. **Per-class counts were reconstructed from the recorded rates and denominators** and are accurate to approximately one row; they are reported as integers because that is how they were reconstructed, not because they were counted. Differences of one or two in a class count carry no weight in any argument here. The two that do carry weight — 49 **Controversial** verdicts falling to 0, and 187 against a base of 49 — are order-of-magnitude effects that survive the uncertainty by a wide margin. §6 states why the underlying per-item outputs cannot be re-derived. An automated gate compares a candidate against a measured baseline on the same lane and precision, and rejects on either an unimproved headline or a per-category floor breach.

**Compute.** Run 1: a single A100, 38,694 rows, one epoch, 2 hours 28 minutes, final training loss 0.0101. The runs are small; the findings are about direction, not scale.

### 3 Run 1 — supervising a three-valued field with binary targets

#### 3.1 The gate rejected it, and was right twice

VERDICT: REJECT (exit 1)

- recall 0.8329 against 0.8554 is within sampling noise at these counts (this set could resolve about 0.036). No improvement was demonstrated, which is not the same finding as a regression.
- per-category recall fell below a measured baseline floor: others 0.7653 below 0.7857; social stereotypes and unfair discrimination 0.6184 below 0.7237

The first clause is a discipline worth naming: **the gate refuses to call an unresolvable difference an improvement**, and says so in the language of what the test set can resolve.

#### 3.2 Against the same-size baseline, the fine-tune made the model worse

|                                | untuned 0.6B | tuned         | $\Delta$       |
|--------------------------------|--------------|---------------|----------------|
| Overall recall                 | 0.8488       | <b>0.8329</b> | <b>-0.0159</b> |
| False-positive rate            | 0.0624       | 0.0519        | -0.0105        |
| Precision                      | 0.9156       | 0.9276        | +0.0120        |
| Controversial verdicts emitted | 49           | <b>0</b>      | extinguished   |

Recall down, false positives down, precision up: **the adapter learned to be more permissive**. For a deny gate that is the wrong direction, and it is the direction that fails silently — a guard that blocks less looks better on every metric an operator watches casually.

**The mechanism is the target vocabulary.** One epoch of binary targets extinguished the third severity class entirely. The rendering function’s own reasoning had been that inventing a **Controversial** target the labels do not distinguish would teach a distinction the data never made, and that downstream policy scoring **Controversial** as harmful “would quietly absorb the difference.” **It did not absorb it.** Every prompt that would have been **Controversial** — and therefore caught, because policy scores it harmful — is now **Safe** and missed.

#### 3.3 The second-order failure: a documented lever became a no-op

This matters more to a governance product than the recall delta.

The deployment exposes an operator policy row: *does a borderline verdict count as harmful?* On the untuned model that lever moves recall from **0.8488 to 0.8011** — a real, documented control with a measurable effect.

On the tuned model, the same lever reports **0.8329, identical to the headline row**, because there are no **Controversial** verdicts left for it to act on.

**A documented operator control silently became inert, and nothing in the system announced it.** No error, no warning, no change in any interface. The lever is still in the

configuration, still in the documentation, and now does nothing. For a system whose purpose is to be auditable, a control that reports success while having no effect is a worse failure than one that breaks loudly.

### 3.4 Targeting the weak category did not fix the weak category

The run’s stated purpose was to improve three underperforming classes. Two moved marginally; the largest collapsed.

| Class (all three targeted)                   | Untuned | Tuned  | $\Delta$ |
|----------------------------------------------|---------|--------|----------|
| fraud_assisting_illegal_activities           | 0.7833  | 0.8000 | +0.0167  |
| others                                       | 0.7551  | 0.7653 | +0.0102  |
| social_stereotypes_and_unfair_discrimination | 0.7661  | 0.6184 | −0.1579  |

The finding is not that LoRA does not work here. It is that **a corpus whose targets are binary trains a binary model**, and weak-category selection does not survive contact with a three-valued output space.

### 3.5 The repair that was not available

Two repairs were on the table. Render **Controversial** for rows the corpora mark borderline, or stop supervising severity at all.

**The first is not implementable from either corpus.** The training mixture’s borderline signal — an agreement column distinguishing unanimous from split annotator judgments — **exists only in the test split**. The train split does not carry it, and the second corpus in the mixture has no equivalent column at all.

That is a corpus-level constraint with a general shape: **the signal needed to supervise a three-valued severity output exists where it cannot be trained on and is absent where it could be**. Any practitioner attempting a graded severity output from these corpora will hit the same wall.

So supervising severity at all means supervising it binary. Which left the second option.

## 4 Run 2 — masking the field, and what the field did

The reasoning was explicit and, we thought, conservative: mask the severity line in the loss, keep it in the input, and the base model’s severity distribution should be left intact while the category line is tuned. Downstream policy treats a gating category as harmful independently, so catch rate could improve through the category path alone.

### 4.1 The measurement

| Run                         | Overall recall | Adversarial recall | Severity verdicts emitted                      |
|-----------------------------|----------------|--------------------|------------------------------------------------|
| Untuned 0.6B                | 0.8488         | 0.7918             | controversial 49, safe 1000, unsafe 650        |
| Run 1 — severity supervised | 0.8329         | 0.7947             | <b>controversial 0</b> , safe 1022, unsafe 677 |

| Run                            | Overall recall | Adversarial recall | Severity verdicts emitted                                                            |
|--------------------------------|----------------|--------------------|--------------------------------------------------------------------------------------|
| <b>Run 2 — severity masked</b> | <b>0.6804</b>  | <b>0.5572</b>      | controversial <b>187</b> , safe 1132, unsafe <b>367</b> , <i>plus category words</i> |

Masking was **17 points of overall recall worse than supervising the field binary, and 23 adversarial points below the untuned model.** The gate rejected run 1 as *within sampling noise* and run 2 as *recall regressed beyond sampling noise* — correctly separating “no improvement shown” from “this made it worse.”

## 4.2 The distributional signature

The recall number says the run failed. The verdict distribution says **why**, and it is the paper’s central evidence.

An isolated field would hold its distribution. Severity did not hold anything:

- **Controversial returned but overproduced** — 187 against the base model’s 49, nearly four times.
- **Unsafe collapsed** — 367 against 650.
- **The model began emitting its own category vocabulary in the severity slot** — strings such as `non-violent illegal acts, others, violent` appearing where `Safe, Unsafe` or `Controversial` belong.

That third observation is the one that settles it. A field that was merely *uncorrected* would drift within its own vocabulary. A field filling with **the neighboring field’s tokens** has not been left alone; it has been actively reshaped by training it was supposedly excluded from.

## 4.3 The mechanism

A LoRA adapter modifies the query, key, value, output, gate, up and down projections across every layer. **Both output fields are produced by those same weights.**

Not computing a loss on severity did not hold severity still. It only stopped *correcting* severity while training moved it anyway. Masking removes a gradient signal; it does not build a wall — and in the absence of a corrective signal, the field drifted toward the token distribution that training was actively reinforcing next door.

Stated generally: **under a shared low-rank update there is no “rest of the model” to hold constant.** Every output field rides the same weights. Loss masking is not a form of parameter isolation and should not be reasoned about as one.

## 5 What follows

**The default was restored to supervising severity binary** — the less bad of two measured losses, and it at least keeps the severity line well-formed. **That is not a fix**, and we record it as a retreat rather than a resolution.

**Isolation requires separate parameters, not a flag.** Three routes remain, and none of them is a third setting of the masking option:

1. **A second adapter** carrying the severity field, with its own parameters.
2. **Target modules that do not carry severity** — which requires knowing which projections produce which field, and is a real open question at this scale.
3. **A corpus that can supervise all three severity values** — which, per §3.5, does not currently exist in the public guard mixtures.

**A note on gates.** Both runs were stopped by an automated parity gate before a human decided anything. The gate’s value here was not that it caught a bad model; it is that it produced two *differently worded* rejections, and the difference between them — unresolvable versus regressed — is what made the second run interpretable as evidence rather than as noise. A promotion gate that emits a single boolean would have discarded the finding.

## 6 Limitations

**Single model, single scale.** One 0.6B model, one adapter configuration. We do not claim the effect size generalizes; we claim the mechanism does, because it follows from weight sharing rather than from any property of this model.

**One adapter configuration tested.** A different rank or a different target-module set might reduce the leakage. **It should not eliminate it**, since the projections producing both fields overlap under every standard configuration we are aware of — but we did not test that, and the claim is therefore mechanistic rather than empirical.

**The raw per-item outputs no longer exist.** The results directory was excluded from version control and was never committed. The figures above survive as recorded prose and as hand-transcribed literals whose per-slice counts were **back-computed from published rates**, and are accurate to approximately one row. **Every number here is reproducible only by re-running the evaluation**, and we state that rather than implying an artifact we cannot ship. This is itself a lesson from the episode, and a more embarrassing one than the failed runs.

**Corpus access.** Both corpora are access-gated, and one carries a commercial-use restriction whose license and gate form disagree. Reproduction requires clearing that.

## 7 Related work

An earlier draft of this section claimed that no treatment existed of what happens to an unsupervised output field when a neighboring field is tuned under a shared adapter. **That claim does not survive a literature check and has been withdrawn.** *Reasoning-Trace Collapse* (arXiv:2605.21127) runs the same configuration — a rank-16 LoRA over the standard attention and MLP projections, four open-weight reasoning models, two-field completions of the form (reasoning, answer), the reasoning field excluded from the cross-entropy but left in the target while the answer field is supervised — and it measures the masked field, which does not hold still: valid-trace rates fall to 61–67 percent and the residual failure mode shifts from empty traces to truncated ones; that work calls masking a partial mitigation. Three narrower components survive, and the contribution below is stated at that width.

Run 1 is the better documented of the two, and is claimed only as a confirming instance. That binary supervision produces binary behavior is standing practice rather than a discovery: the Aegis guardrail datasets carry an explicit third *Needs Caution* label and ship two LoRA-tuned LlamaGuard variants differing only in whether that label was folded into Safe or into Unsafe (Ghosh et al., NAACL 2025). Minority collapse supplies the mechanism — above an imbalance threshold the classifiers for minority classes collapse onto a mutually indistinguishable direction, and a class with no training targets at all is the degenerate case

(Fang et al., PNAS 2021). The result was also predicted in print: Choudhary et al. (COLM 2026) merge a four-level graded harm classifier with a binary refusal model, find graded classification collapsing while refusal transfers under every standard merge rule, and close by naming risk triage and content classification as capabilities vulnerable to being overwritten by a coarser objective. Run 1 confirms that prediction in the more common single-run setting. What the reviewed work does not report is the deployment form of §3.3: an output schema and a documented operator lever that both remain in place after the class they act on has been eliminated.

The mechanism behind run 2 is likewise established, and existing vocabulary is adopted rather than any coined. Multi-task learning names cross-objective degradation through shared parameters *negative transfer*, and its anti-correlated signature the *seesaw phenomenon*, with separation of shared from task-specific parameters as the standard remedy (Tang et al., RecSys 2020). Continual learning gives the sharper form: under a moving backbone, classes receiving no loss term still have their feature distributions drift, which is why *semantic drift* compensation exists (Yu et al., CVPR 2020; Goswami et al., ECCV 2024). The same holds for LoRA specifically — adaptation degrades held-out capability that no loss protects, and the working remedy is an explicit preservation term over the non-target distribution rather than its omission (Xu et al., arXiv:2605.29498) — while interference under a single shared adapter is documented with uniformly parameter-space remedies (arXiv:2504.07448; arXiv:2601.09684). *Parameter isolation* is itself a term of art in continual learning, so the formulation in §4.3 assembles two existing terms and names nothing new. Closest of all, multi-task partially supervised learning makes zeroing an unlabeled task’s loss its standard recipe, and its pseudo-supervision machinery exists because that recipe does not suffice (Kokkinos, CVPR 2017; Li et al., CVPR 2022).

What none of this supplies is the inverse measurement. Masking is evaluated there by its effect on the supervised objective (arXiv:2404.07965; Huerta-Enochian and Ko, EMNLP 2024), or at the granularity of a task, a held-out capability, or a broad disposition (arXiv:2502.17424). No located work reports what the masked positions themselves go on to emit, as a per-class distribution measured against the base model’s. Three claims are therefore advanced: that per-field distributional measurement (**Controversial** 49 to 187, **Unsafe** 650 to 367); the direction of the result, in which masking cost 17 points of recall against supervising the field with targets known to be wrong — the opposite sign to arXiv:2605.21127, whose unmasked baseline supervises a degenerate empty target that masking removes; and the governance consequence, that a masked field is a policy surface whose silent movement invalidates a documented control. The vocabulary migration in §4.2 already has a name — a *placement error*, a correct value emitted at the wrong position, in the taxonomy of Zhang et al. (arXiv:2608.25358) — and that name is adopted here.

The absence asserted above is bounded by how it was tested: three independent English-language sweeps of arXiv, alphaXiv and the open web, covering the PEFT and loss-masking literature, the multi-task interference and partial-supervision literature, and guard-model fine-tuning, with full texts read for the closest hits. No systematic ACL Anthology or citation-graph sweep was run; a forward sweep on arXiv:2605.21127 or the partially-supervised multi-task line could plausibly close the remaining gap, and the claims above are scoped to survive that.

## 8 Conclusion

Masking a field’s loss does not isolate that field under LoRA. The masked field came unmoored, overproduced one class, collapsed another, and filled with its neighbor’s vocabulary — costing 17 points of recall against the alternative it was meant to improve on.

The prior run that motivated the masking is the more useful finding for anyone shipping a

guard: a corpus with binary targets trains a binary model, and it will **extinguish a third output class without announcing it** — silently converting a documented operator control into a no-op that reports success.

We publish both because they cost two runs and a week, the explanations are each one sentence, and we could not find either sentence written down.